Gender, Starting Salary, and Academic Performance of Tepper MBAs

by

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Executive Summary

The objectives of this analysis are twofold. One objective is to investigate whether men and women Tepper MBA graduates with comparable qualifications obtain comparable starting salaries. The other is to investigate whether men and women students with comparable qualifications perform equally well academically in the Tepper MBA program. The data for this analysis are for graduating full-time MBA students in cohorts from 2004 through 2010. The approach uses multiple regression analysis. This widely used method permits investigation of the separate effects of each of several variables that might affect starting salary, and similarly for variables that might affect graduating GPA.

The findings are the following. 1) There is not a significant difference in starting salaries between Tepper men and women MBA graduates who have comparable graduating GPA, work experience, citizenship, and cohort. 2) There is not a significant difference in graduating GPA between Tepper men and women MBA graduates who have comparable GMAT scores, undergraduate GPA, undergraduate area of study, work experience, citizenship, and cohort. The results are also valuable for understanding the importance of qualifications and work experience in affecting starting salary and GPA.

Introduction

Educational institutions such as the Tepper School of Business educate a diverse student body. Ideally, education helps all students reach their potential to have satisfying lives, productive careers, and to make constructive contributions to society. While success in this broad objective is not easily measured, two outcome measures, academic performance and starting salary, are important indicators. The assessment reported here is focused on gender equality with respect to these two outcome measures.

Data

A total of 834 observations are available for the salary analysis reported below. For the analysis of Tepper graduating GPA, data for undergraduate grade point average (UGPA) are not available for the 2004 and 2005 cohorts. For the remaining cohorts, UGPA data are available for 438 of 668 individuals in the sample. Hence, the analysis of graduating GPA (TotalGPA) is conducted using 438 observations.

Salary

To begin, it is informative to look at histograms for Total GPA, salary, and years of work experience. These are shown in Figure 1. From the salary histogram, it is evident that there are several very low salaries: 10 graduates reported a salary of $40,000 or less. The finding reported below (no significant difference in salary by gender) continues to hold if these ten observations are removed from the regression. The finding also continues to hold if the salary outlier on the high end is removed.

The 834 observations for salaries for cohorts 2004 through 2010 are used for the regression in Table 1. The dependent variable is starting salary. The independent variables are graduating GPA, a quadratic in years of work experience, an indicator variable for female, indicator variables for citizenship, and indicator variables for graduation year. The reference year in the regression is 2010. The indicator variable for female has a coefficient of -$1,307. This coefficient is not close to significant—the p-value is .36.

In broader populations, it is often the case that salary data are strongly skewed to the right; there are many more poor people than there are rich people. Hence, regressions using salary data often use the logarithm of salary as the dependent variable. The finding of no significant difference in starting salary between men and women holds if the dependent variable is the logarithm of salary. As evident from Figure 1, however, starting salary data for Tepper graduates do not exhibit skewness. In addition, simple diagnostics favor the linear specification in Table 1 over a specification using the logarithm of salary. Hence, the results from the linear model are reported in Table 1.

Some brief comments on the other variables follow. The graphs in Figure 2 show the estimated variation of salary with work years and GPA. As the left graph shows, the predicted effect of work experience on starting salary peaks at roughly 10 years of pre-MBA work experience. The benefit of 10 years of experience relative to zero experience is estimated to be nearly $16,000. Turning to the coefficient of GPA, we see that a one-point increase in Total GPA is estimated to increase starting salary by $13,554. The coefficients of the graduation date variables in Table 1

are the estimated difference by year relative to year 2010. For example, the estimated average salary gain between 2009 and 2010 was $1,575. The estimated effects of citizenship are relative to a citizen of the US. I have labeled the countries by letter because some countries have few students, and results for those countries may not be very informative. Hence, I decided that it is best not to report the names of individual countries. Only CountryB, CountryF, and CountryG are statistically significant. On average, holding constant other variables in the regression, graduates who are from CountryB are estimated to obtain $6,245 higher starting salaries than US citizens. Graduates who are citizens of CountryF and CountryG are estimated to obtain, respectively, an average of $16,481 and $11,136 less than US citizens. The sample has 100 students who are CountryB, 4 students from CountryF, and 19 from CountryG.

The salary regression accounts for about 20% of variation in starting salary. This suggests, not surprisingly, that starting salaries are determined largely by characteristics of individual graduates that we do not measure. I investigated including indicators for undergraduate degree (engineering, business, quantitative). Those variables have no significant predictive effect. I also included a count of courses taken in each of eleven areas (accounting, communications, economics, entrepreneurship, finance, information systems, marketing, organizational behavior, strategy, quantitative, and production/operations management). These eleven variables increased the R^2 by only .04 to .24 and had negligible impact on the magnitude and significance of the coefficient of female. The coefficient of the indicator variable for female in the latter regression is -$1,321 and the p-value is .38.

With respect to gender, the key message is that there is not a significant difference in starting salaries between Tepper men and women MBA graduates who have comparable graduating GPA, work experience, citizenship, and cohort. This accords with findings by Bertrand, Goldin, and Katz (2010) in their study of University of Chicago graduating MBA classes from 1990 through 2006.[1] They find that male and female graduates have “nearly identical” starting salaries.

GPA of Graduating Students

I next analyze whether there are gender differences in academic performance at Tepper. The regression in Table 2 has GPA at graduation (TotalGPA) as the dependent variable. Data for this analysis are available for cohorts 2006 through 2010.[2] As noted earlier, there are a total of 668 observations for these cohorts, but, due to missing data for undergraduate grade point average (UGPA), only 436 observations are available for the regression. The coefficient of gender is quantitatively small (-.0086) and statistically insignificant (p=.62). Thus, this regression reveals no significant difference between academic performance of men and women in our program. Changing the dependent variable in Table 2 from TotalGPA to GPA in required courses yields the same conclusion with respect to gender. The coefficient on the indicator variable for female is

[1] Marianne Bertrand, Claudia Goldin and Lawrence F. Katz, "Dynamics of the Gender Gap for Young Professionals in the Financial and Corporate Sectors," *American Economic Journal: Applied Economics*, Vol. 2, No. 3 (July 2010), pp. 228-255.

[2] For this regression, I wanted 2010 to be the reference year. To do this, I used the levels command as shown in the R script that accompanies this document.

very small (-.015) and statistically insignificant (p=.50). Given the similarity of findings to the TotalGPA, I do not report the equation for GPA in required courses.

Brief discussion of other variables in Table 2 follows. As expected, the GMAT and undergraduate GPA variables are highly significant. Table 3 expresses the coefficients in standardized form (also known as beta weights). The standardized coefficients for the GMAT and UGPA variables, highlighted in bold, all have a magnitude on the order of .3. Hence, a one standard deviation increase in any one of these three variables increases predicted TotalGPA by approximately .3 standard deviations. Indicator variables for type of undergraduate major are also included in the regression in Table 2: IENG (engineering), quantitative non-engineering major (IQUANT), and business (IBIS). Only the indicator for engineering is significant, with an estimated GPA benefit of .07. The coefficients for graduation date are small and statistically insignificant. This suggests that there was no grade inflation over these five cohorts (which is nice to see). While significant, the effect of years of work experience on GPA is not large. The estimated benefit of work experience peaks at 6 years, with GPA gain of .06 relative to a student with zero experience. For individuals with more than 12 years of experience, the effect of experience on GPA drops below that of a student with zero experience. Only 15 individuals in the sample have more than 12 years of work experience. The reference group for citizenship is USA. The three countries from which students significantly underperform on GPA relative to US students are CountryA, COUNTRYF, and CountryG. The number of these students in the sample is relatively small. There are 10 students from CountryA, 4 students from CountryF, and 19 from CountryG.

Conclusion

The analysis indicates that: 1) There is not a significant difference in starting salaries between Tepper men and women MBA graduates who have comparable graduating GPA, work experience, citizenship, and cohort. 2) There is not a significant difference in graduating GPA between Tepper men and women MBA graduates who have comparable GMAT scores, undergraduate GPA, undergraduate area of study, work experience, citizenship, and cohort.

Figure 1

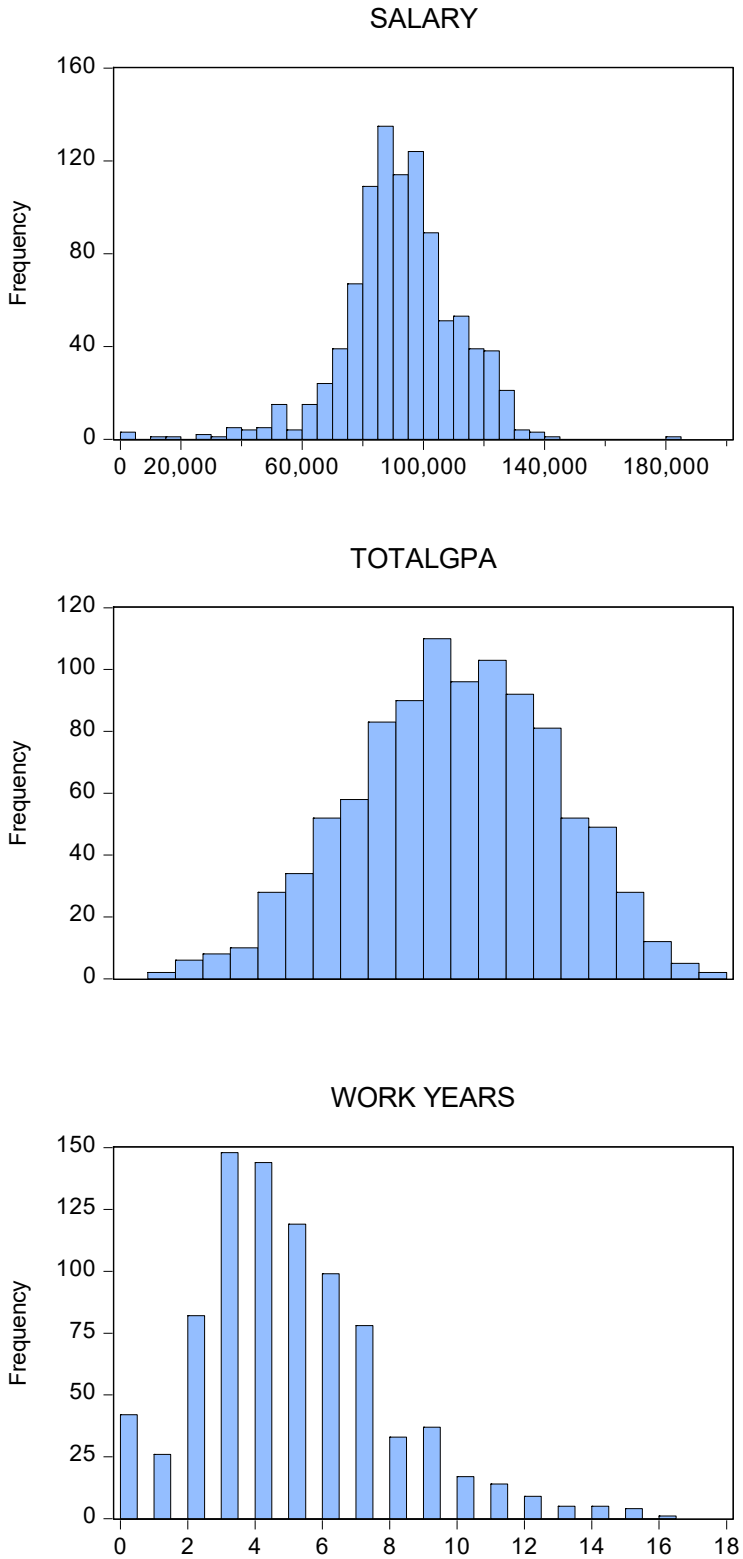
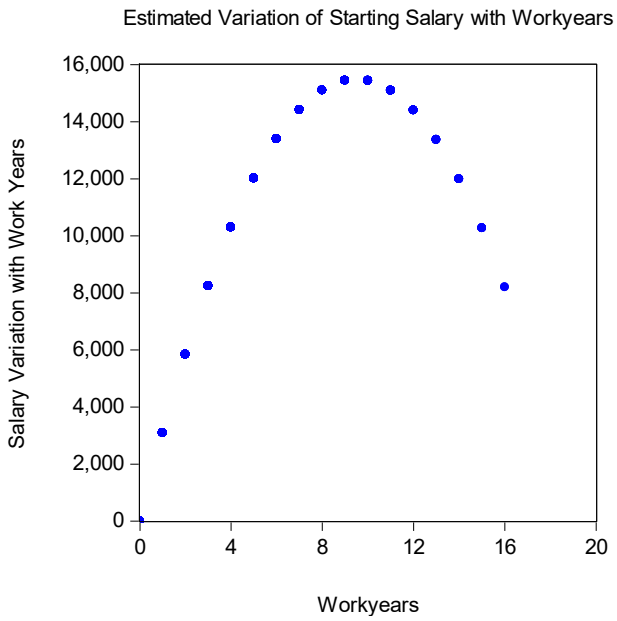

SALARY
Frequency
160
120
80
40
0
0
20,000
60,000
100,000
140,000
180,000
TOTALGPA
Frequency
120
100
80
60
40
20
0
WORK YEARS
Frequency
150
125
100
75
50
25
0
0
2
4
6
8
10
12
14
16
18

Figure 2

The graph on the left shows the gain in starting salary with work experience relative to an individual with zero work experience. The graph on the right shows the gain in starting salary with graduating GPA relative to an individual with GPA of 3.1.



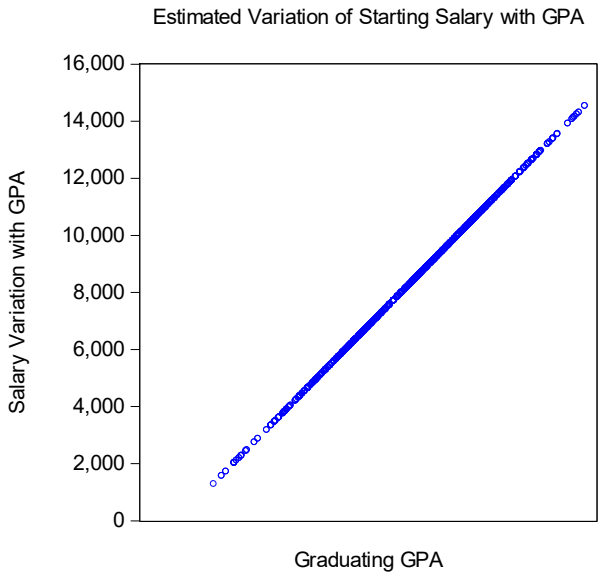


Table 1

Salary Regression

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	31198.396	10801.195	2.8884	0.0039743
TotalGPA	13554.724	2917.021	4.6468	3.930e-06
Workyears	3262.446	679.249	4.8030	1.859e-06
I(Workyears^2)	-171.877	48.369	-3.5535	0.0004021
factor(Gender)F	-1306.978	1435.130	-0.9107	0.3627206
factor(GradYear)2005	-9013.637	1513.423	-5.9558	3.844e-09
factor(GradYear)2006	-1456.089	1675.601	-0.8690	0.3851056
factor(GradYear)2007	3167.935	1704.818	1.8582	0.0634971
factor(GradYear)2008	8614.153	1921.628	4.4827	8.423e-06
factor(GradYear)2009	3376.707	2013.891	1.6767	0.0939827
factor(GradYear)2010	4951.949	1664.566	2.9749	0.0030171
factor(Country)CountryA	1133.288	2842.876	0.3986	0.6902618
factor(Country)CountryB	6244.764	1720.005	3.6307	0.0003003
factor(Country)CountryC	616.664	7875.160	0.0783	0.9376046
factor(Country)CountryD	1767.250	2445.984	0.7225	0.4701873
factor(Country)CountryE	-1519.380	2359.005	-0.6441	0.5197067
factor(Country)CountryF	-16481.536	5265.699	-3.1300	0.0018104
factor(Country)CountryG	-11136.788	5088.321	-2.1887	0.0289017

Residual standard error: 16110 on 816 degrees of freedom
Multiple R-squared: 0.2035, Adjusted R-squared: 0.187
F-statistic: 12.27 on 17 and 816 DF, p-value: < 2.2e-16

Note: Heteroscedasticity-consistent standard errors

Table2

GPA Regression

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Coefficients:
                                                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                                                      1.9204241  0.1161426  16.535  < 2e-16
GMATQuant                                                        0.0143566  0.0019938   7.200 2.85e-12
GMATVerbal                                                       0.0127896  0.0016470   7.766 6.45e-14
GMATWriting                                                      0.0080662  0.0093206   0.865  0.38731
UGPA                                                             0.1560563  0.0178841   8.726  < 2e-16
factor(Gender)M                                                  0.0085740  0.0172238   0.498  0.61889
Workyears                                                        0.0213250  0.0081333   2.622  0.00907
I(Workyears^2)                                                  -0.0017711  0.0006858  -2.583  0.01014
IQUANT                                                           0.0170336  0.0229901   0.741  0.45917
IENG                                                             0.0676164  0.0222383   3.041  0.00251
IBIS                                                             0.0224014  0.0222890   1.005  0.31546
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2006  0.0217349  0.0231916   0.937  0.34921
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2007 -0.0256126  0.0248200  -1.032  0.30270
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2008  0.0039013  0.0239512   0.163  0.87069
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2009  0.0194066  0.0229991   0.844  0.39927
factor(Country)CountryA                                         -0.3025689  0.0436295  -6.935 1.57e-11
factor(Country)CountryB                                         -0.0163150  0.0498764  -0.327  0.74375
factor(Country)CountryC                                         -0.0660518  0.0742772  -0.889  0.37438
factor(Country)CountryD                                         -0.0144877  0.0221835  -0.653  0.51406
factor(Country)CountryE                                          0.0556166  0.0264676   2.101  0.03622
factor(Country)CountryF                                         -0.1288146  0.0410133  -3.141  0.00181
factor(Country)CountryG                                         -0.1918737  0.0387448  -4.952 1.07e-06

Residual standard error: 0.1494 on 414 degrees of freedom
  (221 observations deleted due to missingness)
Multiple R-squared:  0.3921,    Adjusted R-squared:  0.3612
F-statistic: 12.71 on 21 and 414 DF,  p-value: < 2.2e-16

Note: Heteroscedasticity-consistent standard errors
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Table 3

Scaled Coefficients for GPA Regression

	Coefficients	stdcoefs	elasticities
(Intercept)	1.9204	NA	NA
GMATQuant	0.0144	0.3048	0.1764
GMATVerbal	0.0128	0.3153	0.1330
GMATWriting	0.0081	0.0324	0.0107
UGPA	0.1561	0.3404	0.1411
factor(Gender)M	0.0086	0.0188	0.0018
Workyears	0.0213	0.2894	0.0247
I(Workyears^2)	-0.0018	-0.2828	-0.0119
IQUANT	0.0170	0.0387	0.0011
IENG	0.0676	0.1577	0.0046
IBIS	0.0224	0.0549	0.0018
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2006	0.0217	0.0448	0.0011
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2007	-0.0256	-0.0577	-0.0016
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2008	0.0039	0.0080	0.0002
factor(GradYear, levels = c(2010, 2006, 2007, 2008, 2009))2009	0.0194	0.0410	0.0010
factor(Country)CountryA	-0.3026	-0.1725	-0.0009
factor(Country)CountryB	-0.0163	-0.0164	-0.0002
factor(Country)CountryC	-0.0661	-0.0239	-0.0001
factor(Country)CountryD	-0.0145	-0.0243	-0.0004
factor(Country)CountryE	0.0556	0.0860	0.0014
factor(Country)CountryF	-0.1288	-0.0867	-0.0006
factor(Country)CountryG	-0.1919	-0.1611	-0.0013